

Software Engineering – Exam Preparation Notes

1. Introduction

- Software engineering is the systematic and disciplined approach to the development, operation, maintenance, and evolution of software.
- The main goals include quality, reliability, maintainability, cost control, and timely delivery.

2. Software Development Life Cycle

- Typical SDLC activities include requirements analysis, design, implementation, testing, deployment, and maintenance.
- The exact activities and their order depend on the selected development process model.

3. Process Models

- The Waterfall model performs development through sequential phases.
- The Iterative model develops software through repeated cycles.
- Agile development emphasizes short iterations, customer feedback, adaptation, and incremental delivery.

4. Requirements Engineering

- Requirements engineering involves elicitation, analysis, specification, validation, and management of requirements.
- Functional requirements describe what a system should do.
- Non-functional requirements describe qualities or constraints such as performance, security, and usability.

5. Software Design

- Software design converts requirements into an architecture and detailed solution structure.
- Important design concepts include modularity, abstraction, cohesion, and coupling.

6. Software Testing

- Unit testing tests individual components.
- Integration testing checks interactions among components.
- System testing evaluates the complete system.
- Acceptance testing determines whether the software satisfies user or business requirements.

7. Maintenance

- Corrective maintenance fixes defects.
- Adaptive maintenance modifies software for environmental changes.
- Perfective maintenance improves performance or functionality.
- Preventive maintenance reduces the likelihood of future problems.